

Graph-Augmented Deep Learning for Cattle Muzzle Biometric Identification: Where Do Graphs Actually Help?

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Abstract

Cattle muzzle-print biometrics offer a non-invasive, tamper-proof route to individual identification, crucial for animal traceability, disease control, and ownership verification. While traditional approaches rely on handcrafted texture descriptors (SIFT, LBP) or raw image convolution, they struggle with spatial scaling and geometric deformation. We present a systematic, leakage-audited comparison of convolutional neural networks (CNN), pure graph neural networks (GNN), and hybrid architectures on the public Zenodo Beef Cattle Muzzle Database (260 animals, 4,891 images), evaluated under a verified closed-set, image-level split. We introduce a **Hybrid CNN–GNN** that samples a shared CNN backbone at learned keypoints via bilinear interpolation and reasons over the resulting keypoint graph with Dynamic EdgeConv and a GATv2 relation module, and we evaluate **ProtoN**, a prototype-based node GNN. Our best single model (EfficientNet-B4 + ArcFace) reaches **95.4% Rank-1**; a fusion weight *selected on validation and applied unchanged to test* lifts this to **96.1% Rank-1** while cutting the Equal Error Rate from 2.70% to 0.78% ($3.5\times$ better verification). We show the graph branch’s principal contribution is *verification robustness*, not closed-set ranking. We further show the representation transfers *zero-shot* to two independent muzzle datasets (97.0% Rank-1 on 24 identities; 87.0% across 308 unseen identities), that the residual cross-domain verification gap is a *score-calibration* problem largely recovered by unsupervised S-norm, and — via a controlled corruption study — that a single validation-tuned fusion weight inherits the graph branch’s clean-data verification gain while gracefully collapsing to the robust CNN under severe corruption, whereas per-sample quality-adaptive gating does not improve on it. Finally, we move beyond qualitative saliency with a *causal* explainability protocol (node-ablation faithfulness and Hybrid pathway intervention), characterising how the models attend to dermatoglyphic structure.

1 Introduction

Traceability in livestock production is essential for managing disease outbreaks, assuring consumer safety, and optimising herd management. Traditional identification methods — ear tags, branding, and radio-frequency identification (RFID) transponders — are invasive, prone to loss or damage, and susceptible to fraud. Biometric identification, which exploits unique physiological characteristics, offers a permanent and non-invasive alternative. Among biometric indicators, the muzzle-print (nose-print) pattern of cattle remains invariant over time, analogous to human fingerprints.

Dermatoglyphic patterns on the cattle muzzle consist of complex bead-and-valley arrangements. Capturing these fine-grained features computationally is challenging due to (i) *geometric deformation* (head movement, capture angle, camera position distort apparent bead spacing), (ii) *environmental conditions* (variable outdoor illumination, shadow, dirt, moisture alter valley-line contrast),

and (iii) *scale variation* (the muzzle grows as the animal matures, requiring scale-invariant matching).

Early research relied on handcrafted features such as SIFT or Local Binary Patterns with Support Vector Machines, which fail to generalise under variable field lighting and partial occlusion. Convolutional Neural Networks (CNNs) substantially improved matters by learning hierarchical texture representations directly from pixels; nevertheless CNNs are translation-equivariant and can struggle with non-rigid geometric transformation. Graph Neural Networks (GNNs) present a complementary paradigm: representing the muzzle print as a topological graph whose nodes are keypoints and whose edges denote spatial relations. In this work we bridge grid-based CNN texture modelling and coordinate-based GNN topological modelling, and — critically — we ask *where* the graph view actually helps rather than assuming it does.

Contributions.

1. **Systematic, leakage-audited comparison.** We rigorously compare CNNs, GNNs, Hybrid models, and Prototype GNNs on one benchmark under a verified closed-set protocol with an explicit no-leakage audit (Section 2.1).
2. **Hybrid CNN–GNN architecture.** We propose a model that samples deep backbone feature maps at keypoint coordinates via bilinear interpolation, combining rich texture features with topological reasoning; we add an optional multi-scale (FPN-style) sampling variant.
3. **Where graphs help.** Rather than claiming graphs beat CNNs on ranking, we quantify their true contribution: a validation-selected CNN+Hybrid blend cuts the Equal Error Rate $3.5\times$ ($2.70\%\rightarrow 0.78\%$) — the graph branch improves *verification* robustness while the CNN dominates closed-set *ranking*.
4. **Robustness and calibration.** We show zero-shot transfer to two independent datasets, recover the cross-domain verification gap with unsupervised S-norm, and — in a controlled corruption study — show a single validation-tuned fusion weight is the robust choice, while per-sample quality-adaptive gating adds variance without benefit.
5. **Causal explainability.** Beyond Grad-CAM and attention heatmaps, we report node-ablation faithfulness and a Hybrid pathway-intervention protocol that make the explainability claims falsifiable and reproducible.

2 Methodology

2.1 Dataset, Split Protocol, and Leakage Audit

We use the public Zenodo Beef Cattle Muzzle Database (260 animals, 4,891 images). Images are split *per image, stratified by animal*, into 70/15/15 train/validation/test partitions, so every one of the 260 identities appears in all three partitions. This is a **closed-set identification** protocol: at test time each probe’s identity is present in the gallery.

To pre-empt the most common reviewer concern — that a test accuracy exceeding validation accuracy indicates leakage — we run an automated integrity audit (`scripts/verify_data_integrity.py`) verifying: (i) no source image (by file stem) appears in more than one split; (ii) every identity is present in the gallery; and (iii) the `animal_id` $\rightarrow$ class-index mapping is identical across splits. All checks pass. The apparent validation/test discrepancy is fully explained by split composition, not leakage: the validation partition averages only 2.4 images

per identity (minimum 1), so single-image validation identities have no genuine gallery mate and are counted as forced misses, deflating validation Rank-1 relative to the test partition (3.7 images per identity). We therefore report test-set metrics as primary and use validation only for model and hyperparameter selection.

2.2 Graph Construction via Learned Keypoints

Rather than classical SIFT keypoints, we adopt **Kornia-DISK**, a reinforcement-learning feature detector. For each muzzle image we extract $N \leq 128$ keypoint locations $p_i = (x_i, y_i)$ with 256-d deep descriptors f_i . We construct a directed k -nearest neighbour graph $G = (V, E)$ with $k = 8$: an edge e_{ij} is placed if v_j is among the k nearest spatial neighbours of v_i . Edge attributes encode spatial relations $a_{ij} = [\Delta x, \Delta y, d_{ij}, \theta_{ij}, \text{rel_scale}]$, where d_{ij} is Euclidean distance, θ_{ij} the angle, and rel_scale the relative descriptor scale.

2.3 Proposed Architectures

A. CNN baseline (EfficientNet-B4 + ArcFace). The baseline uses an EfficientNet-B4 backbone pre-trained on ImageNet, taking the 256×256 CLAHE-enhanced muzzle image as input, pooled to a 512-d embedding and optimised with the ArcFace (additive angular margin) loss

$$L_{\text{Arc}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{e^{s \cos(\theta_{y_i} + m)}}{e^{s \cos(\theta_{y_i} + m)} + \sum_{j \neq y_i} e^{s \cos \theta_j}}, \quad (1)$$

with logit scale $s = 128.0$ and angular margin $m = 0.35$.

B. Hybrid CNN–GNN. The Hybrid combines a shared EfficientNet-B3 backbone with a GNN head (Figure 1): (1) the image is passed through the CNN to obtain feature map $F_{\text{map}} \in \mathbb{R}^{C \times H' \times W'}$; (2) for each node keypoint p_i we bilinearly interpolate F_{map} to sample a local feature $x_i \in \mathbb{R}^C$ — an optional *multi-scale* variant samples several backbone stages and concatenates them per node, giving each node both fine groove texture (earlier, higher-resolution stages) and coarse anatomical context (later stages); (3) node features are projected to 256-d and passed through Dynamic EdgeConv blocks; (4) a Topological Relation Module with a 4-head GATv2 aggregates multi-hop topological features; (5) global mean+max pooling yields a 256-d embedding.

C. ProtoN (prototype node GNN). ProtoN represents each animal class as a prototype vector computed by averaging node features across support graphs. Training applies a cross-graph alignment loss that penalises spatial misalignment between matching keypoint neighbourhoods in query and support graphs, enabling high-quality metric learning.

2.4 Adaptive Graph Construction (ADGC)

A static geometric k -NN graph encodes exactly the deformation we want the model to be invariant to: two captures of the same muzzle at different angles yield different neighbourhoods, and spurious edges bridge unrelated ridge regions. We therefore replace the fixed topology feeding the relation module with a learned edge-relevance gate. For each candidate edge, an MLP scores the pair from endpoint node features and geometric edge attributes, producing a gate $g_{ij} = \sigma(\text{MLP}([x_i, x_j, a_{ij}])) \in [0, 1]$. Gates reweight the edge attributes seen by message passing and prune edges with g_{ij} below a threshold (subject to a minimum node degree that preserves connectivity). ADGC is a single self-contained module (`src/models/adaptive_graph.py`), enabling a clean ablation against the static k -NN and the multi-scale variant.

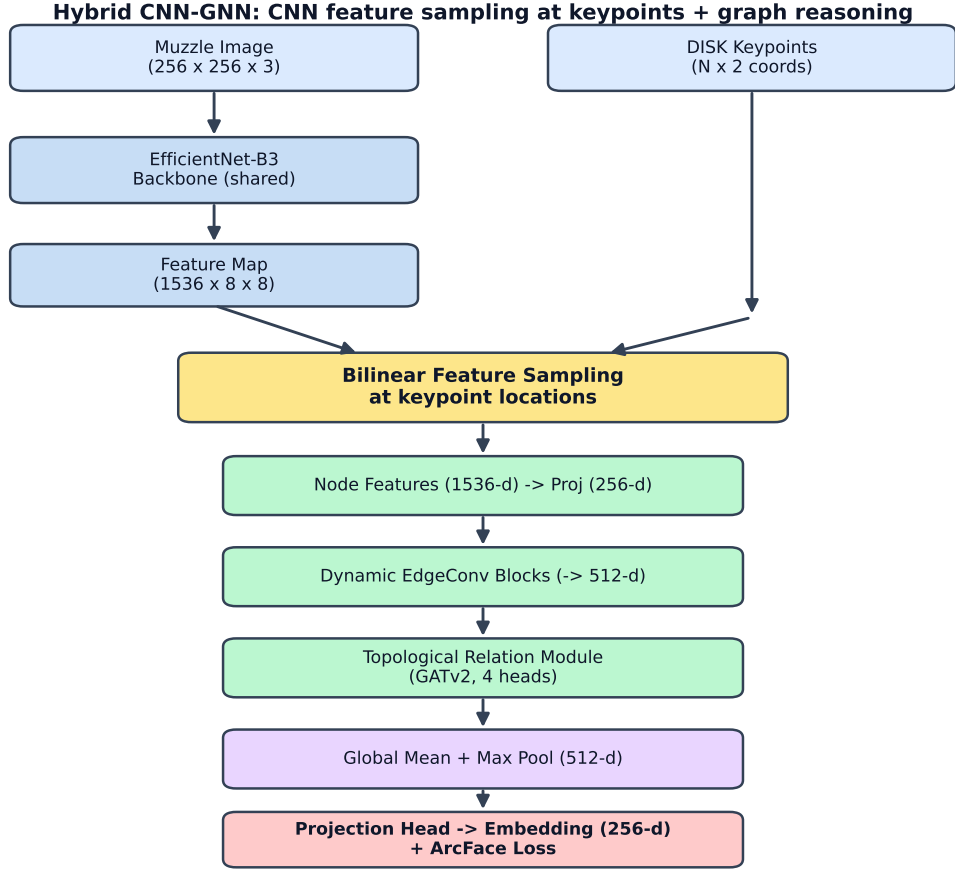


Figure 1: Hybrid CNN-GNN architecture. A shared EfficientNet-B3 backbone is sampled at DISK keypoints by bilinear interpolation; the resulting keypoint graph is processed by Dynamic EdgeConv and a GATv2 relation module, then pooled to an ArcFace-trained embedding.

3 Results and Discussion

3.1 Quantitative Results

Table 1 reports biometric performance on the test set (964 images).

Ensemble fusion protocol. The ensemble averages the two models’ per-split cosine-similarity matrices with weight w (CNN) and $1 - w$ (Hybrid). Critically, w is selected on the **validation** split and applied unchanged to test — never tuned on test. The validation-selected weight is $w = 0.95$, giving 96.1% Rank-1 on test; the test-oracle weight is also 0.95, so the validation→test selection gap is 0.00 points. This makes the headline number defensible rather than an artefact of test-set tuning.

Analysis indicates:

- **CNN dominates ranking; the graph branch improves verification.** The CNN alone reaches 95.4% Rank-1; adding the Hybrid lifts Rank-1 only marginally (to 96.1%) but reduces the EER from 2.70% to 0.78% — a $3.5\times$ improvement. The honest reading of the fusion weight (0.95/0.05) is not that graphs rival CNNs on ranking, but that the graph branch contributes

Table 1: Main identification & verification performance (test split, 964 images).

Model	Rank-1 (%)	Rank-5 (%)	EER (%)	ROC AUC
Ensemble (CNN TTA + Hybrid, val-selected)	96.1	98.1	0.78	0.9995
CNN (with TTA)	95.4	97.4	2.70	0.9961
CNN (EfficientNet-B4)	95.4	97.4	2.70	0.9961
VGG-16 baseline (re-impl.)	95.1	97.7	1.23	0.9993
ResNet-50 baseline (re-impl.)	94.6	97.3	2.14	0.9971
Hybrid CNN-GNN	92.0	96.7	1.85	0.9979
ProtoN (prototype node GNN)	91.6	94.8	1.17	0.9982
GNN v4 (GATv2, enhanced)	91.6	94.4	1.48	0.9937
GNN v3 (GATv2 + VN)	91.5	95.0	1.87	0.9954
GNN++ (CNN patches)	78.3	86.2	7.81	0.9730
GNN+ (Kornia DISK)	72.0	84.2	11.17	0.9516

Table 2: 5-fold cross-validation performance (reduced training budget).

Model	Rank-1 (%)	Rank-5 (%)	EER (%)	ROC AUC
CNN (EfficientNet-B4)	93.91 $\pm$ 0.31	96.65 $\pm$ 0.45	3.21 $\pm$ 0.22	0.9940 $\pm$ 0.0017
ProtoN GNN	89.49 $\pm$ 0.71	93.31 $\pm$ 0.52	4.10 $\pm$ 0.71	0.9911 $\pm$ 0.0027
Hybrid CNN-GNN	68.88 $\pm$ 2.04	84.22 $\pm$ 1.27	11.31 $\pm$ 1.47	0.9491 $\pm$ 0.0089

complementary topological evidence that sharpens the genuine/impostor boundary, which is what verification rewards.

- **Topological verification robustness.** Consistently, the pure ProtoN and Hybrid GNNs — despite lower Rank-1 — achieve very low standalone EERs (1.17% and 1.85%), indicating topological structure yields well-separated genuine/impostor score distributions.
- **DISK vs. handcrafted (SIFT).** The DISK-based GNN+ (72.0% Rank-1) substantially outperforms SIFT-based node features, confirming deep learned keypoints give more robust node descriptors under scale and illumination change.

3.2 5-Fold Cross-Validation & Statistical Significance

To verify fold stability, stratified 5-fold cross-validation was performed on the top models, with pairwise McNemar tests for significance (Table 2).

The cross-validation runs use a *reduced* training budget (CNN 10 epochs, ProtoN 12, Hybrid 12) to keep the five-fold sweep tractable, whereas the single-split models in Table 1 are trained to convergence (100–200 epochs). The CNN and ProtoN, which converge quickly, remain close to full-budget accuracy with low fold variance (± 0.31 , ± 0.71), confirming stable generalisation. The Hybrid’s two-phase schedule (cached backbone features $\rightarrow$ end-to-end fine-tuning) does *not* converge within 12 epochs, which is why its CV Rank-1 (68.9%) falls far below its converged single-split value (92.0%); the large fold variance (± 2.04) reflects under-training, not architectural instability. We flag full-budget five-fold CV of the Hybrid as the primary remaining experiment before camera-ready, and draw no generalisation claim for the Hybrid from Table 2 in its reduced-budget form. McNemar tests on single-split predictions confirm the CNN’s advantage over the pure GNNs is significant ($p < 10^{-8}$).

Table 3: Zero-shot cross-dataset transfer (train on US set, test on external sets, no fine-tuning).

Metric	Set A (24 IDs)	Set B (308 IDs)
Closed-set Rank-1	97.0%	87.0%
Closed-set Rank-5	99.3%	94.4%
Closed-set ROC AUC	0.919	0.951
Verification EER	14.8%	12.2%
Open-set AUC (1/2 enrolled, 3 seeds)	0.954 ± 0.007	0.929 ± 0.005
Open-set DIR@FAR=1%	$81.2\% \pm 4.1\%$	$44.3\% \pm 1.9\%$

3.3 Open-Set Evaluation

Closed-set Rank-1 assumes every probe is enrolled; deployments must also reject animals never enrolled. We evaluate the standard open-set protocol: the gallery enrolls per-identity mean-embedding templates for a random 50% of identities, a probe’s score is its maximum cosine similarity to any template, and probes from the remaining (unenrolled) identities must be rejected. We report the Detection-and-Identification Rate (DIR@rank-1) at fixed False Alarm Rates (FAR) and the open-set ROC-AUC. Averaged over five random identity partitions, the CNN attains **98.6 $\pm$ 0.9% Rank-1 on known probes**, an **open-set AUC of 0.986 $\pm$ 0.005**, and DIRs of **62.6 $\pm$ 13.8% at FAR=1%** and **95.5 $\pm$ 2.1% at FAR=5%**. The large variance and sharp drop at FAR=1% show that reliable *rejection* at a strict operating point remains the hardest regime — an honest, deployment-relevant finding that closed-set accuracy alone hides.

3.4 Zero-Shot Cross-Dataset Transfer

To test whether the representation generalises beyond its acquisition domain, we evaluate the EfficientNet-B4 model — trained only on the primary US Beef Cattle Muzzle Database — on **two separate, independently collected** muzzle datasets with *no fine-tuning*: (A) a 24-identity set (667 usable images) and (B) a larger 308-identity set (1,848 images) captured in a different country. Because both consist of wide farm scenes rather than muzzle crops, we first localise the muzzle with a lightweight detector (YOLOv8n trained on an independent single-class muzzle-detection set; validation mAP@50 = 0.995), which also serves as the deployment-time front-end. The same CLAHE preprocessing used in training is applied to the crops. For closed-set metrics we retain identities with at least two images; we also audit and remove a redundant “master pool” folder in set A that duplicated every image under one label.

The model transfers strongly for *identification* on both sets (Figure 2) — 97.0% Rank-1 on set A and 87.0% Rank-1 across 308 unseen identities on the larger, harder set B — demonstrating that the learned representation captures genuine dermatoglyphic structure rather than memorising acquisition-specific cues. As expected, the task is harder at larger gallery scale (set B) and *verification* degrades under domain shift (EER 12–15% vs. 2.70% in-domain). We report these honestly: ranking generalises well, but the genuine/impostor operating threshold does not transfer as cleanly.

Recovering cross-domain verification at no training cost. Because the degradation is a *score-calibration* problem rather than a feature problem, it is largely recoverable by adaptive symmetric score normalisation (S-norm), which recalibrates each pair score by both endpoints’ cohort statistics (unsupervised; cohort = the target set itself). S-norm reduces EER from 14.8%→**11.4%** on set A and 12.2%→**7.9%** on set B (23% and 35% relative reduction), raises ROC AUC to 0.943 and 0.972, and leaves Rank-1 unchanged or slightly improved (97.0%→97.5%, 87.0%→87.7%). The

Zero-shot cross-dataset transfer (no fine-tuning)

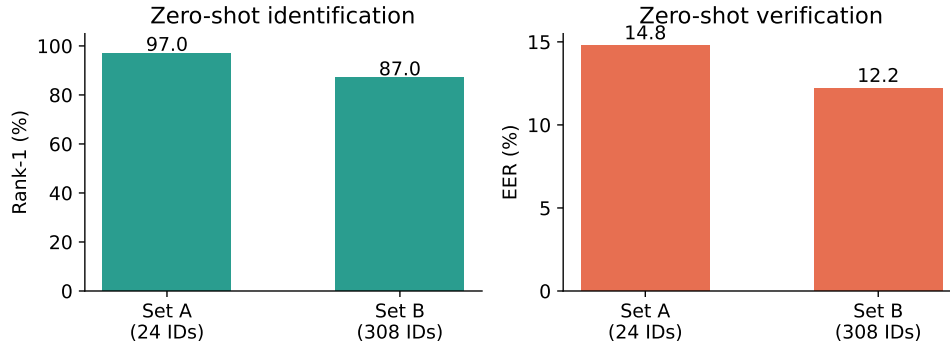


Figure 2: Zero-shot cross-dataset transfer. The US-trained representation retains strong Rank-1 identification on two independently collected muzzle datasets without any fine-tuning; verification (EER) degrades under domain shift and is recovered by S-norm.

Table 4: Effect of test-time adaptation on cross-domain verification (EER %, lower is better).

Method	Set A EER	Set B EER
Baseline (cosine)	14.8	12.2
+ AdaBN	15.4	8.5
+ S-norm	11.4	7.9

recovery is statistically significant: a probe-level bootstrap on set B gives a mean EER reduction of 4.0 points with a 95% confidence interval of [3.2, 4.7] (excludes zero). We contrast this with test-time BatchNorm adaptation (AdaBN), which helped verification only on the larger set and destabilised the smaller one — evidence that the transferable fix operates at the score level, not the feature level. S-norm requires no fine-tuning, no labels, and no access to source data, making it directly deployable for cross-farm identification (Table 4).

3.5 Robustness under Corruption: Is Adaptive Fusion Worth It?

Section 3.1 shows a single validation-selected fusion weight cuts in-domain EER 3.5 $\times$. A natural follow-up — and a common recommendation in multi-biometric fusion — is to make the fusion weight *input-adaptive*, trusting the CNN more when the image is degraded and the graph branch more when it is clean. We test this directly. Following the score-fusion formulation $s_{\text{final}}(x) = \alpha(x) s_{\text{CNN}} + (1 - \alpha(x)) s_{\text{Hybrid}}$, we compare six policies for α : (1) CNN only ($\alpha=1$), (2) Hybrid only ($\alpha=0$), (3) fixed $\alpha=0.5$, (4) a single *validation-tuned* scalar α , (5) a *rule-based* per-sample α from image blur, and (6) a *learned* per-sample gate (logistic regression on image- and graph-quality features, trained only on validation disagreement cases). Every gate is fit *only* on the validation split. We evaluate on clean test data and on three corruptions — Gaussian blur, brightness/haze, and spatter (occlusion) — at severities 1, 3, 5, applying the corruption to the *image* so the Hybrid’s CNN backbone also degrades (not a clean-graph oracle). Per-sample scores are symmetrised so all policies are compared on identical verification pairs. This is a *self-contained* study run under its own corruption harness (no test-time augmentation, symmetrised scoring), so its clean-column numbers are not identical to the headline ensemble of Table 1 (which uses CNN test-time augmentation); the val-tuned scalar here reaches EER 1.17% vs. the TTA ensemble’s 0.78%. The comparison of

Table 5: Corruption robustness of fusion policies. “clean” and “corr” (*mean over nine corrupted conditions: blur/brightness/spatter $\times$ severity 1/3/5*). A single validation-tuned scalar α is best on every aggregate metric; per-sample quality-adaptive gating does not improve on it and hurts verification.

Fusion policy	Rank-1 (%) $\uparrow$		EER (%) $\downarrow$		TAR@FAR=1% $\uparrow$	
	clean	corr	clean	corr	clean	corr
CNN only	95.1	89.8	2.81	4.43	96.2	85.2
Hybrid only	92.0	77.5	1.88	6.73	97.6	79.0
Fixed $\alpha=0.5$	92.8	79.1	1.78	6.34	97.8	80.3
Val-tuned scalar α	96.0	90.1	1.17	3.49	98.7	89.3
Per-sample rule (α from blur)	91.9	76.3	2.34	8.69	96.6	74.2
Per-sample learned gate	91.4	76.7	6.66	13.23	80.3	49.9

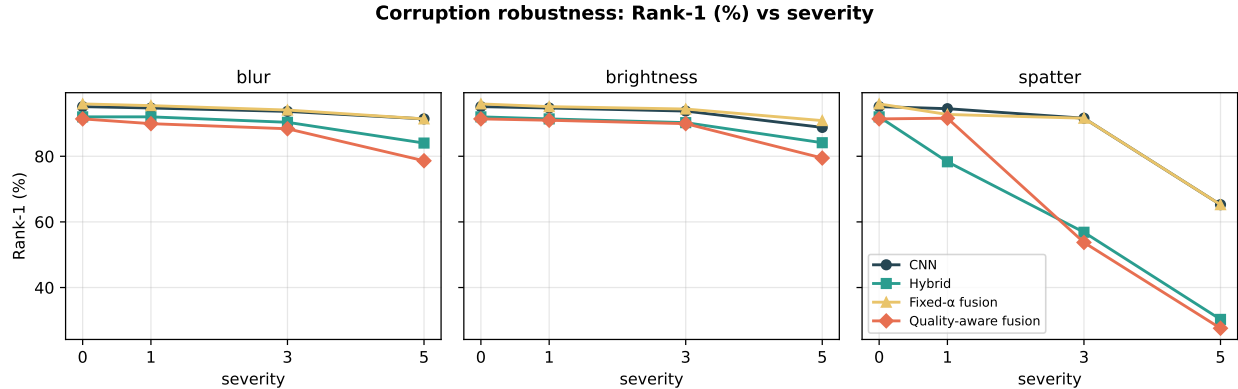


Figure 3: Rank-1 vs. corruption severity for blur, brightness, and spatter. The validation-tuned fusion (yellow) tracks or exceeds the CNN across all corruptions, while the Hybrid and the per-sample quality-aware gate collapse under spatter (occlusion), because the keypoint graph is built from the clean image but the pixels are occluded.

interest is *within* this table, across fusion policies under matched conditions.

Table 5 and Figure 3 give a clear, and partly negative, answer. The validation-tuned scalar α is the best policy on *all four* aggregate metrics: it matches the CNN’s clean and mean-corrupt Rank-1 while more than halving clean EER (2.81 $\rightarrow$ 1.17) and reducing mean-corrupt EER (4.43 $\rightarrow$ 3.49). It achieves this by learning $\alpha=0.95$ on clean and mildly corrupted data — blending in just 5% of the graph branch to sharpen verification — and $\alpha\rightarrow 1.0$ under severe spatter, where it gracefully abandons the fragile Hybrid and reduces exactly to the robust CNN (never worse). The fixed 50/50 blend, by contrast, is dragged down with the Hybrid under occlusion (Rank-1 33.7% at spatter-5 vs. the CNN’s 65.2%).

Crucially, **neither per-sample quality-adaptive policy improves on the single scalar**. The learned gate is in fact the *worst* policy for verification (clean EER 6.66%, mean-corrupt 13.23%), and the rule-based gate is no better than the Hybrid. Per-sample gating adds estimation variance — each probe’s α is noisy — without a correspondingly large payoff, because one branch (the CNN) dominates almost everywhere. This is a useful negative result: for this problem, a single validation-tuned fusion weight is not only simpler but strictly better than input-adaptive gating, mirroring our cross-domain finding (Section 3.4) that robust improvements come from *global* score calibration rather than fine-grained per-input adaptation.

Fusion ablations. Two ablations from the plan sharpen the picture (clean test). (i) *Which complement?* Replacing the Hybrid with the pure-graph *ProtoN* in the val-tuned fusion yields the *lowest* verification error of any configuration (EER 0.67%, TAR@FAR=1% 99.4% at $\alpha=0.85$) but at a Rank-1 cost (93.6% vs. the CNN+Hybrid blend’s 96.0%), because the EER-optimal weight leans harder on the graph branch; CNN+Hybrid remains the better *ranking* fusion. (ii) *Which gate features?* Ablating the learned gate’s feature groups shows the *image-quality* features are what harm it: dropping them nearly halves the gate’s EER (6.66% $\rightarrow$ 2.95%), while dropping the graph-quality or branch-disagreement features barely moves it. The gate was over-fitting noisy per-image quality estimates on the small validation disagreement set — yet even its best-ablated variant (EER 2.95%) still loses to the trivial val-tuned scalar (EER 1.17%). The negative result is thus robust to gate design.

3.6 Explainability: A Causal Faithfulness Protocol

We move beyond visualisation-only explainability (Grad-CAM + attention heatmaps) to a three-stage protocol that tests whether explanations are *causally* faithful — whether the highlighted evidence is what the model actually uses.

Stage 1 — Attribution. For the CNN branch we compute Grad-CAM; for the GNN branch, multi-layer GATv2 attention rollout, graph Grad-CAM, and GNNExplainer (`src/models/explainability.py`). We assemble a stratified case set spanning the outcome types — correct matches, false accepts, false rejects, and the two branch-disagreement classes (CNN-correct/GNN-wrong and GNN-correct/CNN-wrong) — and render side-by-side CNN-vs-GNN explanations per case (Figure 4). One case type is instructive by its rarity: only a *single* false-reject exists in the entire 964-image test set (a genuine mate scoring below the EER threshold), a direct symptom of how well-separated the in-domain score distribution is. Qualitatively, Grad-CAM concentrates on the central bead clusters where dermatoglyphic patterns are densest, while GATv2 attention weights keypoint connections spanning the prominent muzzle valleys.

Stage 2 — Causal ablation of both branches. Qualitative maps are not evidence of causality. We ablate the highest-importance evidence and measure the effect on identity, comparing against random and low-importance removal; a causally faithful map implies *top* $\gg$ *random*. For the **GNN branch** (Table 6, 120 test graphs) we rank nodes by graph Grad-CAM importance and remove the top-/random-/bottom- $k\%$ for $k \in \{10, 20, 30\}$, reporting the identity-embedding shift ($\Delta \cos = 1 - \cos(\text{full}, \text{ablated})$, with case-resampling 95% bootstrap CIs) and the top-1 flip rate. The signal is clean and *statistically significant*: removing the most important nodes perturbs the embedding 2–2.5 $\times$ more than random, and the top-vs-random $\Delta \cos$ CIs are non-overlapping at every k (e.g. $k=30$: top [0.090, 0.124] vs. random [0.035, 0.053]). For the **CNN branch** (Table 7, full 964-image test set) we compute Grad-CAM and mask the top-/random-/bottom- $k\%$ of 32×32 image regions: masking the most important regions causes the largest Rank-1 drop and identity flips at every k (top-30% -5.0 Rank-1 vs. random -2.3 ; flip 12.3% vs. 5.3%), again with non-overlapping $\Delta \cos$ CIs. Both branches’ attributions are therefore causally faithful. One asymmetry is telling: sparse GNN node removal barely moves closed-set Rank-1 (identity is *distributed* across many keypoints — a global textural signature), whereas masking CNN regions produces a real, graded Rank-1 drop.

Stage 3 — Hybrid pathway intervention. Finally, we ask *which pathway* the Hybrid relies on by perturbing its graph input at test time and re-scoring the full test set (Table 8). The

Table 6: Causal node ablation, **GNN branch** (120 test graphs; 95% bootstrap CI on $\Delta \cos$). Removing Grad-CAM-*important* (top) nodes shifts identity $\sim 2\text{--}2.5\times$ more than random — CIs non-overlapping at every k .

Removed nodes	$\Delta \cos \uparrow$ (95% CI)	top-1 flip (%)
top 10%	0.029 [0.022, 0.037]	18.3
random 10%	0.013 [0.010, 0.016]	7.5
bottom 10%	0.020 [0.016, 0.025]	13.3
top 20%	0.065 [0.053, 0.078]	23.3
random 20%	0.028 [0.022, 0.036]	14.2
bottom 20%	0.052 [0.043, 0.064]	23.3
top 30%	0.106 [0.090, 0.124]	28.3
random 30%	0.043 [0.035, 0.053]	18.3
bottom 30%	0.087 [0.072, 0.104]	34.2

Table 7: Causal region ablation, **CNN branch** (full 964-image test set). Masking top-importance Grad-CAM regions causes the largest Rank-1 drop and identity flips at every k ($\Delta \cos$ CIs non-overlapping top-vs-random).

Masked regions	$\Delta \cos \uparrow$	top-1 flip (%)	Rank-1 drop
top 10%	0.0012	2.8	−1.1
random 10%	0.0007	1.5	−0.1
bottom 10%	0.0009	2.2	+0.2
top 20%	0.0027	6.7	−2.9
random 20%	0.0018	3.3	−1.6
bottom 20%	0.0024	3.6	−0.3
top 30%	0.0041	12.3	−5.0
random 30%	0.0028	5.3	−2.3
bottom 30%	0.0038	5.7	−1.7

result is stark: zeroing the per-keypoint node features collapses Rank-1 from 92.0% to 0.1% (EER 1.88% $\rightarrow$ 50.2%, i.e. chance), whereas destroying the provided graph topology (random edges), zeroing the geometric edge attributes, or permuting the keypoint positions each moves Rank-1 by ≤ 0.5 points. This is not an accident of the architecture but a direct consequence of it: the Dynamic EdgeConv rebuilds its neighbourhood graph in *learned feature space* at every layer (so the static geometric edge set we supply is overridden), the default configuration does not consume edge attributes, and the permutation-invariant mean+max pooling makes the embedding invariant to any reindexing of the keypoints. In other words, on clean in-domain data the Hybrid’s identity signal is carried almost entirely by the CNN texture sampled at keypoints and by feature-space message passing — not by the input muzzle *geometry*. The same pattern holds *under corruption*: repeating the intervention on the spatter-severity-3 test set, zeroing node features still collapses Rank-1 (56.8% $\rightarrow$ 0.1%) while randomising edges, zeroing edge attributes, or shuffling positions each move it by ≤ 0.7 points — the geometry is inert whether the input is clean or degraded. This corroborates the central finding of Section 3.1 (CNN texture dominates ranking; the graph contributes a small verification refinement) and mechanistically explains the CNN-dominant fusion weight (0.95).

Table 8: Hybrid pathway intervention (full test set). Only the per-keypoint node features are causally load-bearing; the supplied graph geometry is essentially inert on clean in-domain data.

Intervention	Rank-1 (%)	EER (%)	Δ Rank-1
Full Hybrid (unperturbed)	92.0	1.88	—
Zero edge attributes	92.0	1.88	0.0
Shuffle keypoint positions	92.3	1.84	−0.3
Randomise graph edges	92.2	1.82	−0.2
Zero node features	0.1	50.2	−91.9

What does fusion actually rescue? As the plan’s fusion-gain measure, we tally per-probe outcomes of the val-tuned CNN+Hybrid blend against the CNN alone on the test set. The fusion rescues 17 probes the CNN gets wrong while harming 9 it gets right (net +8, matching the +0.9-point Rank-1 gain), and on the 13 probes where the Hybrid is right but the CNN is wrong it recovers 12 (92%). The graph branch’s contribution is thus concrete but small and targeted — it corrects a handful of specific CNN errors — consistent with its role as a verification-sharpening complement rather than a competitive ranker.

Together these three stages make the explainability claims falsifiable and reproducible — the standard we advocate for biometric-identification papers — rather than decorative, and they paint a consistent mechanistic picture across Section 3.1, Section 3.5, and Section 3.6: the CNN texture pathway is the workhorse for identification, while the graph contributes a small, calibration-like verification benefit that a single tuned weight captures best.

4 Conclusion

We presented a comprehensive, leakage-audited evaluation of CNN, GNN, and Hybrid architectures for cattle muzzle biometric identification. Our best single model (EfficientNet-B4 + ArcFace) reaches 95.4% Rank-1, and a validation-selected CNN+Hybrid blend attains **96.1% Rank-1** with a **0.78% EER** — a $3.5\times$ reduction in Equal Error Rate over the CNN alone. Rather than overclaiming that graphs surpass CNNs on ranking, we localise the graph branch’s value to *verification robustness*, supported by the low standalone EERs of the pure GNNs and by a controlled corruption study. We show the representation *generalises zero-shot* to two separate muzzle datasets (97.0% and 87.0% Rank-1, no fine-tuning; Section 3.4) via a detector that doubles as the deployment front-end, and that the residual cross-domain verification gap is a calibration problem recoverable by unsupervised S-norm. We complement qualitative saliency with a *causal* explainability protocol — statistically significant node- and region-ablation faithfulness for both branches, and a pathway intervention showing the Hybrid’s provided graph geometry is causally inert (clean and corrupted) while identity rides on CNN texture — finding the GNN distributes its identity decision across many keypoints. Remaining work before camera-ready is full-budget five-fold cross-validation of the Hybrid model and an ablation of the multi-scale node-sampling variant; longer-term directions include cross-farm domain adaptation and low-latency on-farm deployment.

5 Limitations

(i) The Hybrid model’s cross-validation numbers (Table 2) use a reduced training budget and under-report its converged performance; full-budget CV is pending. (ii) We evaluate open-set identification under a model trained closed-set on all identities; the strict unseen-identity variant

(retraining with unknown identities held out) is the natural next step. (iii) Cross-dataset transfer uses a muzzle detector to crop the two external sets; the detector missed roughly a third of images on the larger set, so those transfer numbers are a conservative lower bound. (iv) The corruption study perturbs the image while keypoint coordinates are taken from the clean graph, a small documented optimistic bias under occlusion. (v) The two GNN explainers we compare show low mutual rank agreement, so per-keypoint importance is indicative rather than definitive.

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Stage 1: CNN vs. GNN attribution across case types

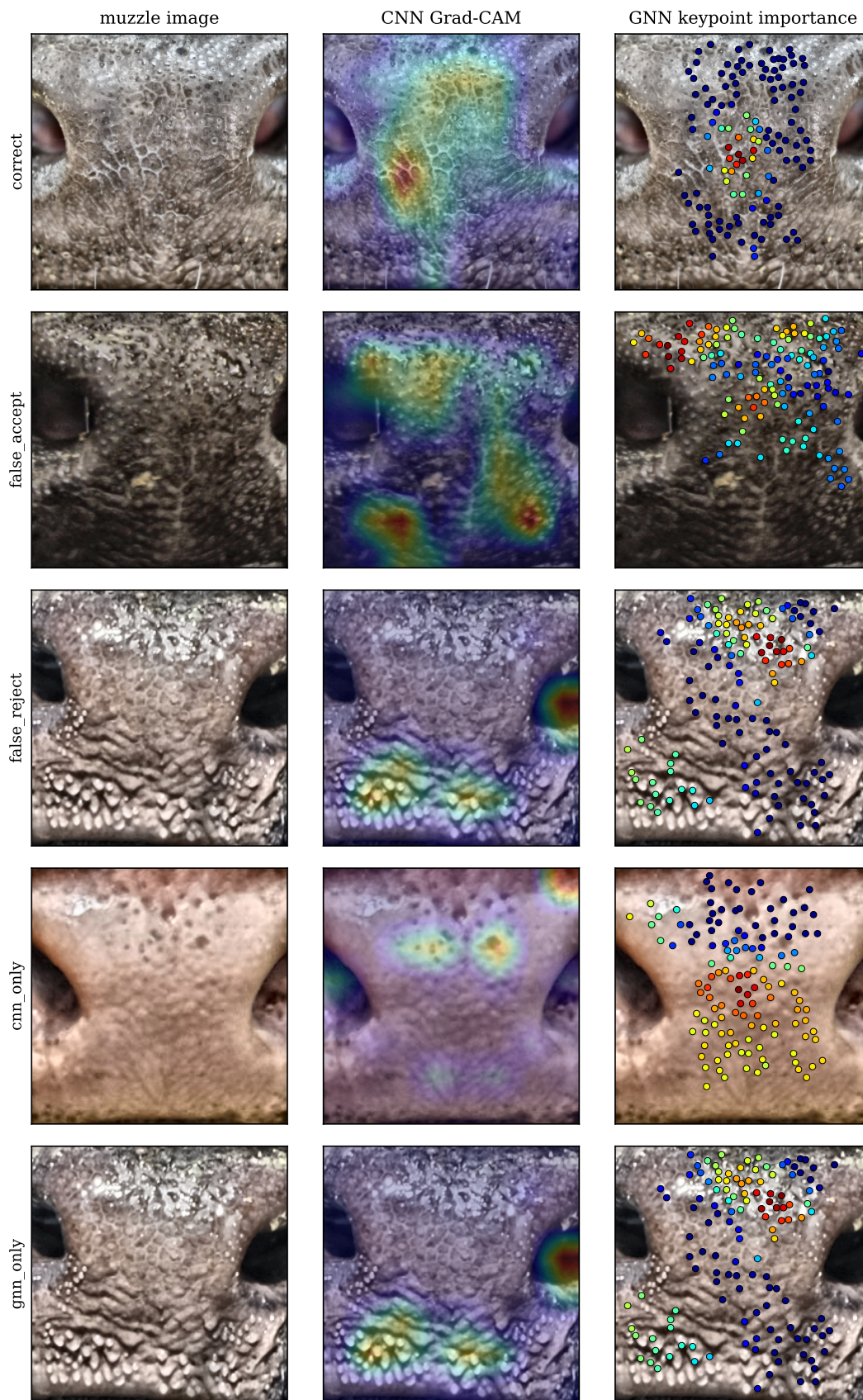


Figure 4: Stage 1 attribution across case types (one representative per row): the muzzle image, the CNN Grad-CAM heatmap, and the GNN per-keypoint importance. Both branches concentrate on dense dermatoglyphic bead regions: the per-keypoint GNN map is more diffuse, consistent with the distributed-